

OPERATORS

Operators:

An operator is a special symbol that operates on data.

Types of Operators:

- 1) Unary Operators
- 2) Binary Operators
- 3) Ternary Operators

Unary Operators:

An operator that operates on only one operand is called as unary operator.

Examples: +a, -a, not a, ~a,.. etc.,

Binary Operators:

An operator that operates on two operands is called as binary operator.

Examples: a+b, a-b, a*b, a/b, a//b, a%b, a**b, a<b, a>b, a<=b, a>=b, a==b, a!=b, .. etc.,

Ternary Operators:

An operator that operates on three operands is called as ternary operator.

No ternary operator in python

Operators are divided into the following categories based on their operations:

- 1) Arithmetic Operators
- 2) Relational Operators
- 3) Logical Operators
- 4) Bitwise Operators
- 5) Assignment Operators
- 6) Membership Operators

7) Identity Operators

8) Unary Plus & Unary Minus Operators

1) Arithmetic Operators:

<u>Operator</u>	<u>Meaning</u>
+	Addition
-	Subtraction
*	Multiplication
/	Division
//	Floor Division
%	Modulo Division
**	Exponentiation

Division operator returns quotient in float format whereas floor division operator returns quotient in int format.

Modulo division operator returns remainder.

2) Relational Operators:

<u>Operator</u>	<u>Meaning</u>
<	Less than
>	Greater than
<=	Less than or equals to
>=	Greater than or equals to
==	Equals to
!=	Not equals to

The above all operators returns bool type value.

3) Logical Operators:

<u>Operator</u>	<u>Meaning</u>
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and	Logical AND
or	Logical OR
not	Logical NOT

Logical AND operator returns true if both the expressions returns true, otherwise returns false.

Logical OR operator returns false if both the expressions returns false, otherwise returns true.

Logical NOT operators reverse the logical state.

4) Bitwise Operators:

<u>Operator</u>	<u>Meaning</u>
&	Bitwise AND
	Bitwise OR
^	Bitwise XOR
<<	Left Shift
>>	Right Shift
~	tilde

Note: All bitwise operators operate on binary data.

5) Assignment Operators:

<u>Operator</u>	<u>Meaning</u>
=	Normal Assignment
a+=b	a=a+b
a-=b	a=a-b
a*=b	a=a*b
a/=b	a=a/b
a//=b	a=a//b

<code>a%=b</code>	<code>a=a%b</code>
<code>a**=b</code>	<code>a=a**b</code>

The above 5 operators are called arithmetic assignment operators.

<code>a&=b</code>	<code>a=a&b</code>
<code>a =b</code>	<code>a=a b</code>
<code>a^=b</code>	<code>a=a^b</code>
<code>a<<=b</code>	<code>a=a<<b</code>
<code>a>>=b</code>	<code>a=a>>b</code>

The above 5 operators are called bitwise assignment operators.

6) Membership Operators:

Membership operators are used to search for the required element in the given iterable object.

<u>Operator</u>	<u>Meaning</u>
in	It returns true if the given element is present, otherwise returns false.
not in	It returns true if the given element is not present, otherwise returns false.

7) Identity Operators:

Identity Operators are used to compare the memory locations of two objects.

<u>Operator</u>	<u>Meaning</u>
is	It returns true if both the operands points to the same object, otherwise returns false.
is not	It returns true if both the operands do not point to the same object, otherwise returns false.

8) Unary Plus & Unary Minus Operators:

<u>Operator</u>	<u>Meaning</u>
+	Unary Plus
-	Unary Minus

By

Mr. Venatesh Mansani

Naresh i Technologies